Lab 3: Component Modelling & Architectural Pattern Selection

WRITTEN JUSTIFICATION

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Assigned System: Public Bus Live Tracking & Crowding Estimator

Architecture Selection

I chose Microservices Architecture for the Public Bus Live Tracking & Crowding Estimator System.

Reason 1: Independent Scaling for High-Volume Live Telemetry

The system must support telemetry from up to 1,000 active buses while GPS data is received every 5 seconds. Separating GPS & Telemetry, ETA, Crowding and Alert functions into independently deployable services allows the telemetry-processing service to be scaled without scaling every other function. This is well suited to the system's real-time and high-volume workload.

Reason 2: Fault Isolation and Independent Service Operation

The system performs different tasks such as receiving GPS data, calculating ETA, estimating crowding and sending service alerts. With microservices, these functions are separated so a problem in one service is less likely to stop the entire tracking system. Services can also be updated or maintained independently, which is useful for a continuously operating public transport system.

Security Advantage

The architecture keeps external access behind service APIs, allowing authentication, role-based authorization and TLS to be applied to requests and telemetry connections. This supports the requirement that unauthenticated requests are rejected and that commuter accounts cannot access Fleet Controller functions. The separation also prevents the commuter interface from directly accessing the transit database.

Performance Benefit

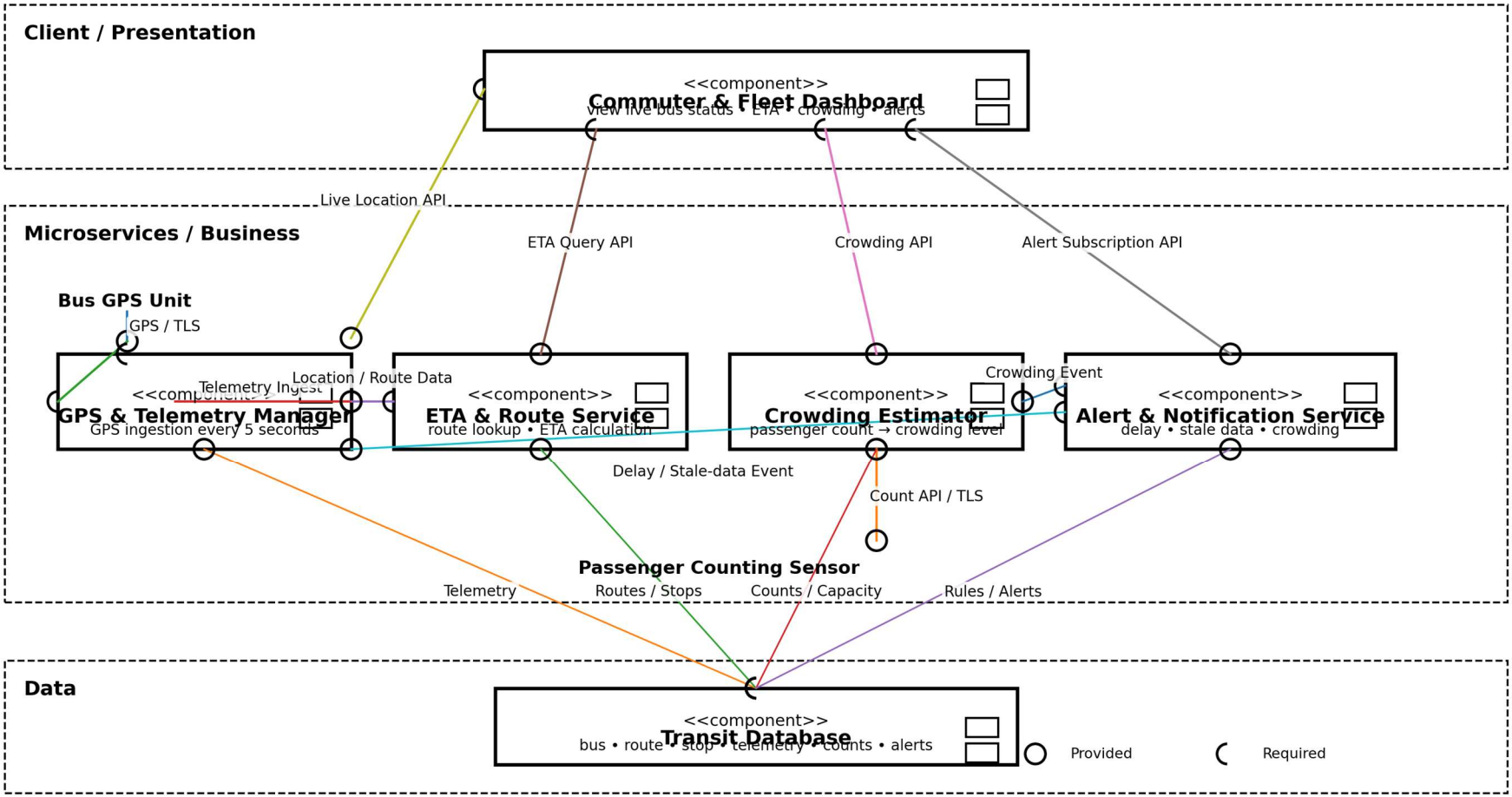
Independent scaling allows the busiest service, especially GPS & Telemetry Manager, to receive additional processing capacity without scaling unrelated services. This helps the system maintain the required performance when many buses are active at the same time and supports the requirement that processing should remain within the specified latency limits.

Identified Components

1. Commuter & Fleet Dashboard Component — Client / Presentation
2. GPS & Telemetry Manager Component — Service
3. ETA & Route Service Component — Service
4. Crowding Estimator Component — Service
5. Alert & Notification Service Component — Service
6. Transit Database Component — Data

Component Diagram

Public Bus Live Tracking & Crowding Estimator — Microservices Architecture



Solid connectors show service communication; ball = provided interface, socket = required interface.

UML interfaces and service interactions are shown explicitly to satisfy the Lab 3 component-modelling requirements.